# Data Schema Design

SYNCOS PRODUCT CASE: ROOT CAUSE ANALYSIS

## Table 1: Task_Execution_Log

**Granularity:** 1 row per completed task lifecycle.
**Primary Use:** Validating H1 (Empty Travel), H2 (Station Bottlenecks), and H5 (Task Asymmetry).

| Column Name | Data Type | Description |
|---|---|---|
| `Task_ID` | String | Unique identifier for the task. |
| `Task_Type` | Enum | Classification of payload (Pallet/Material). |
| `Robot_ID` | String | Specific vehicle assigned to task. |
| `Time_Created` | Timestamp | When the WMS/ERP requested the task. |
| `Time_Assigned` | Timestamp | When SyncOS dispatched a specific robot. |
| `Time_Robot_Arrived_Pick` | Timestamp | When robot physically reached origin. |
| `Time_Pick_Complete` | Timestamp | When payload was successfully loaded. |
| `Time_Drop_Complete` | Timestamp | When payload was successfully delivered. |

## Table 2: Fleet_State_Log

**Granularity:** 1 row per state-change event (or high-frequency time tick) per robot.
**Primary Use:** Validating H3 (Staging Deadlocks), H4 (Battery Logic), and H6 (Environmental Micro-Delays).

| Column Name | Data Type | Description |
|---|---|---|
| `Log_ID` | String | Unique identifier for the telemetry tick. |
| `Timestamp` | Timestamp | Exact time of the recorded state. |
| `Robot_ID` | String | Specific vehicle recording the state. |
| `Current_State` | Enum | Moving, Blocked_Traffic, Charging, Fault. |
| `Location_Zone` | String | Logical area (e.g., Aisle_4, Charging_Bay). |
| `Battery_Level_Pct` | Integer | Current battery charge remaining (0-100). |